Project 2 – TMA4300

Karine Grande & Thea Boge

2026-03-02

```
## Warning in check_dep_version(dep_pkg = "TMB"): package version mismatch:
## glmmTMB was built with TMB package version 1.9.17
## Current TMB package version is 1.9.18
## Please re-install glmmTMB from source or restore original 'TMB' package (see '?reinstalling' for more
```

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## Warning in attr(.knitEnv$meta, "knit_meta_id"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
## Warning in attr(.knitEnv$meta, "knit_meta_id"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
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```

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## Warning in attr(.knitEnv$meta, "knit_meta_id"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Problem 1

We suppose that the joint density of U, Z up to a normalizing constant is

$$f^*_{U,Z}(u,z) = \begin{cases} 1, & \text{for } 0 < u < e^{-z^2/2}, \\ 0, & \text{elsewhere.} \end{cases}$$

a)

If a joint density is known up to proportionality, the full conditional of one variable is obtained by

$$f_{X|Y}(x|y) \propto f^*_{U,Z}(u,z) \textit{as a function of } x$$

Conditional ditribution of $U|Z = z$

Fix z. The joint density becomes

$$f^*_{U,Z}(u,z) = \begin{cases} 1, & \text{for } 0 < u < e^{-z^2/2}, \\ 0, & \text{elsewhere.} \end{cases}$$

As a function of u, this is constant on the interval $(0, e^{-z^2/2})$.

The normalizing constant is

$$\int_0^{e^{-z^2/2}} 1\, du = e^{-z^2/2}.$$

Therefore,

$$f_{U|Z}(u \mid z) = \frac{1}{e^{-z^2/2}} = e^{z^2/2}, \quad 0 < u < e^{-z^2/2}.$$

Hence,

$$U \mid Z = z \sim \text{Unif}\big(0, e^{-z^2/2}\big).$$

Conditional ditribution of $Z|U = u$

Fix u. The support condition is $0 < u < e^{-z^2/2}$.

Solving for z:

$$u < e^{-z^2/2} \quad \Longleftrightarrow \quad \log u < -\frac{z^2}{2}$$

$$-2\log u > z^2$$

$$|z| < \sqrt{-2\log u}$$

Define

$$a(u) = \sqrt{-2\log u}$$

Therefore, the support of the conditional distribution is

$$z \in (-a(u), a(u)).$$

Within this interval the joint density is constant and equal to 1. The normalizing constant is therefore the length of the interval:

$$\int_{-a(u)}^{a(u)} 1\, dz = 2a(u).$$

Thus, the full conditional density of $Z \mid U = u$ is

$$f_{Z|U}(z \mid u) = \frac{1}{2a(u)}, \quad |z| < a(u),$$

where

$$a(u) = \sqrt{-2 \log u}.$$

Thus,

$$Z \mid U = u \sim \text{Unif}\big(-a(u), a(u)\big).$$

Gibbs Sampler

Given an initial valid point $(U^{(0)}, Z^{(0)})$ satisfying $0 < U^{(0)} < e^{-(Z^{(0)})^2/2}$, iterate for $t = 1, \ldots, n$:

1. Sample
$$U^{(t)} \sim \text{Unif}\left(0, e^{-(Z^{(t-1)})^2/2}\right).$$

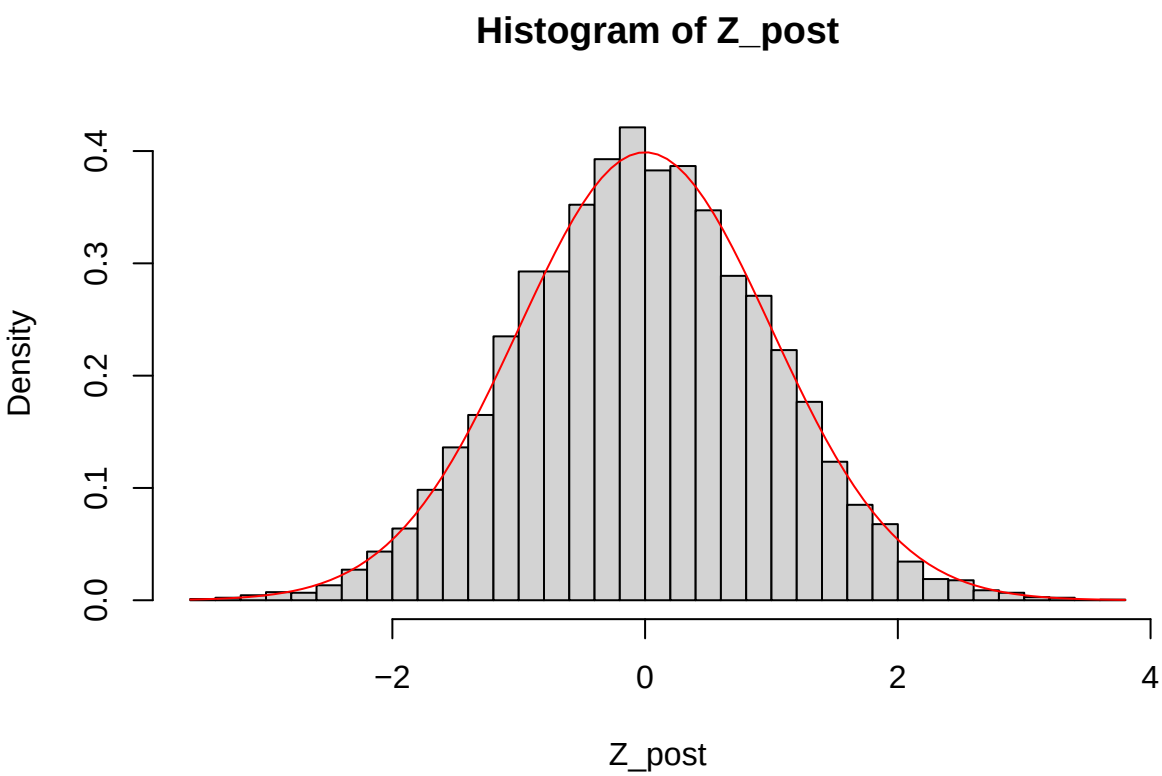
2. Define
$$a^{(t)} = \sqrt{-2 \log U^{(t)}}.$$

3. Sample
$$Z^{(t)} \sim \text{Unif}\left(-a^{(t)},\, a^{(t)}\right).$$

This iterative scheme defines a Markov chain whose stationary distribution is the joint density $f_{U,Z}(u, z)$.The marginal stationary distribution of Z is $\mathcal{N}(0, 1)$.

```
## [1] -0.01857042
```

```
## [1] 0.9699554
```



The posterior sample mean is approximately

$$\hat{\mu} \approx -0.019,$$

and the empirical variance is

$$\hat{\sigma}^2 \approx 0.97.$$

Both values are close to the theoretical mean and variance,

$$\mathbb{E}[Z] = 0, \qquad \mathrm{Var}(Z) = 1,$$

of a standard normal distribution.

The histogram aligns well with the standard normal density curve, confirming that the Gibbs sampler correctly targets

$$Z \sim \mathcal{N}(0, 1).$$

b)

We determine the marginal distribution of Z by integrating out U.

Recall that the joint density (up to normalization) is

$$f^*_{U,Z}(u, z) = \begin{cases} 1, & 0 < u < e^{-z^2/2}, \\ 0, & \text{otherwise.} \end{cases}$$

The marginal density of Z is therefore

$$f_Z(z) = \int_0^{e^{-z^2/2}} 1 \, du = e^{-z^2/2}.$$

Thus,

$$f_Z(z) \propto e^{-z^2/2}.$$

Since

$$\int_{-\infty}^{\infty} e^{-z^2/2}\, dz = \sqrt{2\pi},$$

the normalized density is

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

Therefore,

$$Z \sim \mathcal{N}(0,1).$$

Hence, discarding the U-values from the Gibbs sampler yields a sample from a standard normal distribution.

c)

Correlation between successive Gibbs samples Z_{n-1} and Z_n}

From the Gibbs sampler we have

$$Z_n \mid U_n \sim \mathrm{Unif}\big(-a(U_n), a(U_n)\big), \qquad a(u) = \sqrt{-2\log u}.$$

This conditional distribution is symmetric around zero. Therefore,

$$\mathbb{E}[Z_n \mid U_n] = 0.$$

By the law of total expectation,

$$\mathbb{E}[Z_n] = \mathbb{E}[\mathbb{E}[Z_n \mid U_n]] = \mathbb{E}[0] = 0.$$

Since the Gibbs sampler has stationary distribution $Z_n \sim \mathcal{N}(0,1)$, it follows that

$$\mathbb{E}[Z_{n-1}] = \mathbb{E}[Z_n] = 0.$$

We now compute the covariance. By definition,

$$\mathrm{Cov}(Z_{n-1}, Z_n) = \mathbb{E}[Z_{n-1}Z_n] - \mathbb{E}[Z_{n-1}]\mathbb{E}[Z_n].$$

Because both expectations are zero, this reduces to

$$\mathrm{Cov}(Z_{n-1}, Z_n) = \mathbb{E}[Z_{n-1}Z_n].$$

Using the tower property (law of iterated expectation),

$$\mathbb{E}[Z_{n-1}Z_n] = \mathbb{E}[\mathbb{E}[Z_{n-1}Z_n \mid U_n]]\,.$$

In the Gibbs sampler, Z_n is generated from U_n independently of Z_{n-1}. Hence, conditional on U_n, the variables Z_{n-1} and Z_n are independent. Therefore,

$$\mathbb{E}[Z_{n-1}Z_n \mid U_n] = Z_{n-1}\,\mathbb{E}[Z_n \mid U_n].$$

Since $\mathbb{E}[Z_n \mid U_n] = 0$, we obtain

$$\mathbb{E}[Z_{n-1}Z_n \mid U_n] = 0.$$

Taking expectations again,

$$\mathbb{E}[Z_{n-1}Z_n] = 0.$$

Thus,

$$\mathrm{Cov}(Z_{n-1}, Z_n) = 0.$$

Successive Gibbs samples Z_{n-1} and Z_n are uncorrelated in stationarity. However, they are not independent, since both depend on the intermediate variable U_n.

Correlation between successive second moments Z_{n-1}^2 and Z_n^2

Recall that

$$Z_n \mid U_n \sim \mathrm{Unif}\big(-a(U_n), a(U_n)\big), \qquad a(u) = \sqrt{-2\log u}.$$

For a symmetric uniform distribution on $(-a, a)$ we have

$$\mathbb{E}[X^2] = \frac{a^2}{3}.$$

Therefore,

$$\mathbb{E}[Z_n^2 \mid U_n] = \frac{a(U_n)^2}{3} = \frac{-2\log U_n}{3}.$$

Unlike the first moment case, this conditional second moment depends explicitly on U_n.

We now compute the covariance using the law of total covariance:

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathbb{E}\big[\mathrm{Cov}(Z_{n-1}^2, Z_n^2 \mid U_n)\big] + \mathrm{Cov}\big(\mathbb{E}[Z_{n-1}^2 \mid U_n], \mathbb{E}[Z_n^2 \mid U_n]\big).$$

In the Gibbs sampler, Z_n is generated from U_n independently of Z_{n-1}. Hence, conditional on U_n, the variables Z_{n-1} and Z_n are independent, and therefore

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2 \mid U_n) = 0.$$

Thus,

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathrm{Cov}\big(\mathbb{E}[Z_{n-1}^2 \mid U_n], \mathbb{E}[Z_n^2 \mid U_n]\big).$$

From the previous calculation,

$$\mathbb{E}[Z_{n-1}^2 \mid U_n] = \mathbb{E}[Z_n^2 \mid U_n] = \frac{-2\log U_n}{3}.$$

Therefore,

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathrm{Var}\left(\frac{-2\log U_n}{3}\right).$$

To evaluate this variance, we determine the marginal distribution of U_n in stationarity.
It can be shown that

$$T = -2\log U_n \sim \chi_3^2.$$

Hence,

$$\mathbb{E}[T] = 3, \qquad \mathrm{Var}(T) = 6.$$

It follows that

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathrm{Var}(T/3) = \frac{6}{9} = \frac{2}{3}.$$

Successive second moments are positively correlated:

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \frac{2}{3} \neq 0.$$

Thus, although successive Gibbs samples Z_{n-1} and Z_n are uncorrelated, their squared values exhibit positive correlation due to the shared dependence on the intermediate variable U_n.

Problem 2

a)

```
##  Family: gaussian  ( identity )
## Formula:          y ~ (1 | i)
## Data: data
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     280.3     288.1    -137.2     274.3        97
##
## Random effects:
##
## Conditional model:
##  Groups   Name        Variance Std.Dev.
##  i        (Intercept) 0.2301   0.4797
##  Residual             0.7560   0.8695
## Number of obs: 100, groups:  i, 20
##
## Dispersion estimate for gaussian family (sigma^2): 0.756
##
## Conditional model:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  10.1888     0.1381   73.79   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The maximum likelihood estimate of the overall mean is $\hat{\mu} = 10.1888$. The estimated variance of the random intercepts is $\widehat{\mathrm{Var}}(\gamma_i) = 0.2301$, corresponding to $1/\nu$, and the estimated residual variance is $\widehat{\mathrm{Var}}(\epsilon_{ij}) = 0.7560$, corresponding to $1/\tau$. These estimates will later be used to check that the Gibbs sampler implementation produces reasonable results.

b) Full conditional distributions

From the hierarchical model

$$y_{ij} \mid \mu, \gamma_i, \tau \sim N(\mu + \gamma_i, \tau^{-1}),$$

$$\gamma_i \mid \nu \sim N(0, \nu^{-1}),$$

with priors

$$\pi(\mu) \propto 1, \qquad \tau \sim \mathrm{Gamma}(\alpha, \alpha), \qquad \nu \sim \mathrm{Gamma}(\alpha, \alpha),$$

we derive the following full conditional distributions.

Let $N = \sum_{i=1}^{m} n_i$.

Full conditional of μ Combining the likelihood with the flat prior for μ, we obtain

$$\mu \mid \gamma, \tau, y \sim N(\bar{y}_\gamma, (N\tau)^{-1}),$$

where

$$\bar{y}_\gamma = \frac{1}{N} \sum_{i=1}^{m} \sum_{j=1}^{n_i} (y_{ij} - \gamma_i).$$

Full conditional of γ_i Using the likelihood contribution from cluster i together with the prior $\gamma_i \sim N(0, \nu^{-1})$, we obtain

$$\gamma_i \mid \mu, \tau, \nu, y \sim N(m_i, (\nu + n_i\tau)^{-1}),$$

where

$$m_i = \frac{\tau \sum_{j=1}^{n_i} (y_{ij} - \mu)}{\nu + n_i\tau}.$$

Full conditional of τ Let

$$S = \sum_{i=1}^{m} \sum_{j=1}^{n_i} (y_{ij} - \mu - \gamma_i)^2.$$

Combining the Gaussian likelihood with the prior $\tau \sim \text{Gamma}(\alpha, \alpha)$ gives

$$\tau \mid \mu, \gamma, \nu, y \sim \text{Gamma}\left(\alpha + \frac{N}{2}, \alpha + \frac{S}{2}\right).$$

Full conditional of ν Using the prior for γ_i and $\nu \sim \text{Gamma}(\alpha, \alpha)$, we obtain

$$\nu \mid \mu, \gamma, \tau, y \sim \text{Gamma}\left(\alpha + \frac{m}{2}, \alpha + \frac{1}{2} \sum_{i=1}^{m} \gamma_i^2\right).$$

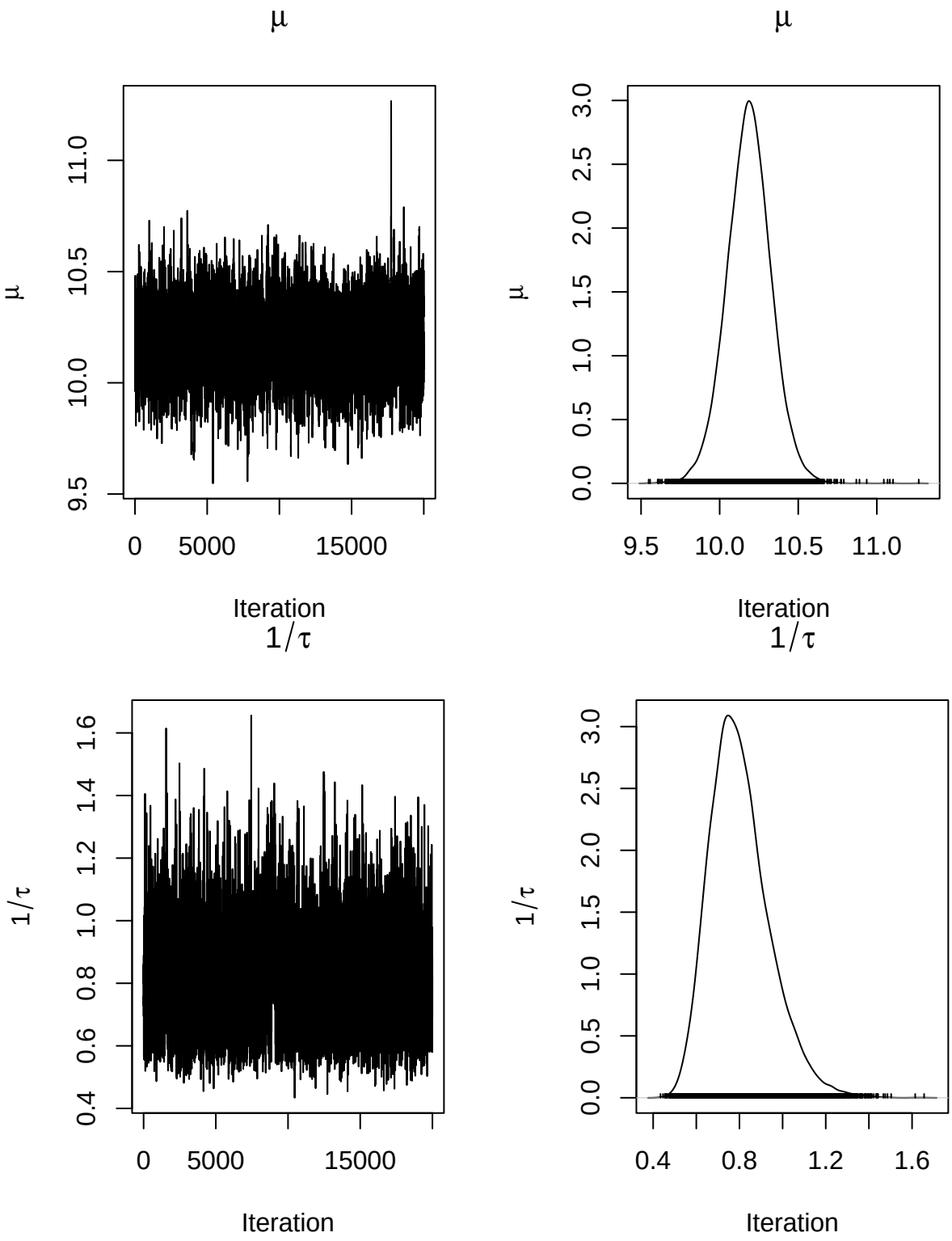
c) Implementation of a Gibbs sampler

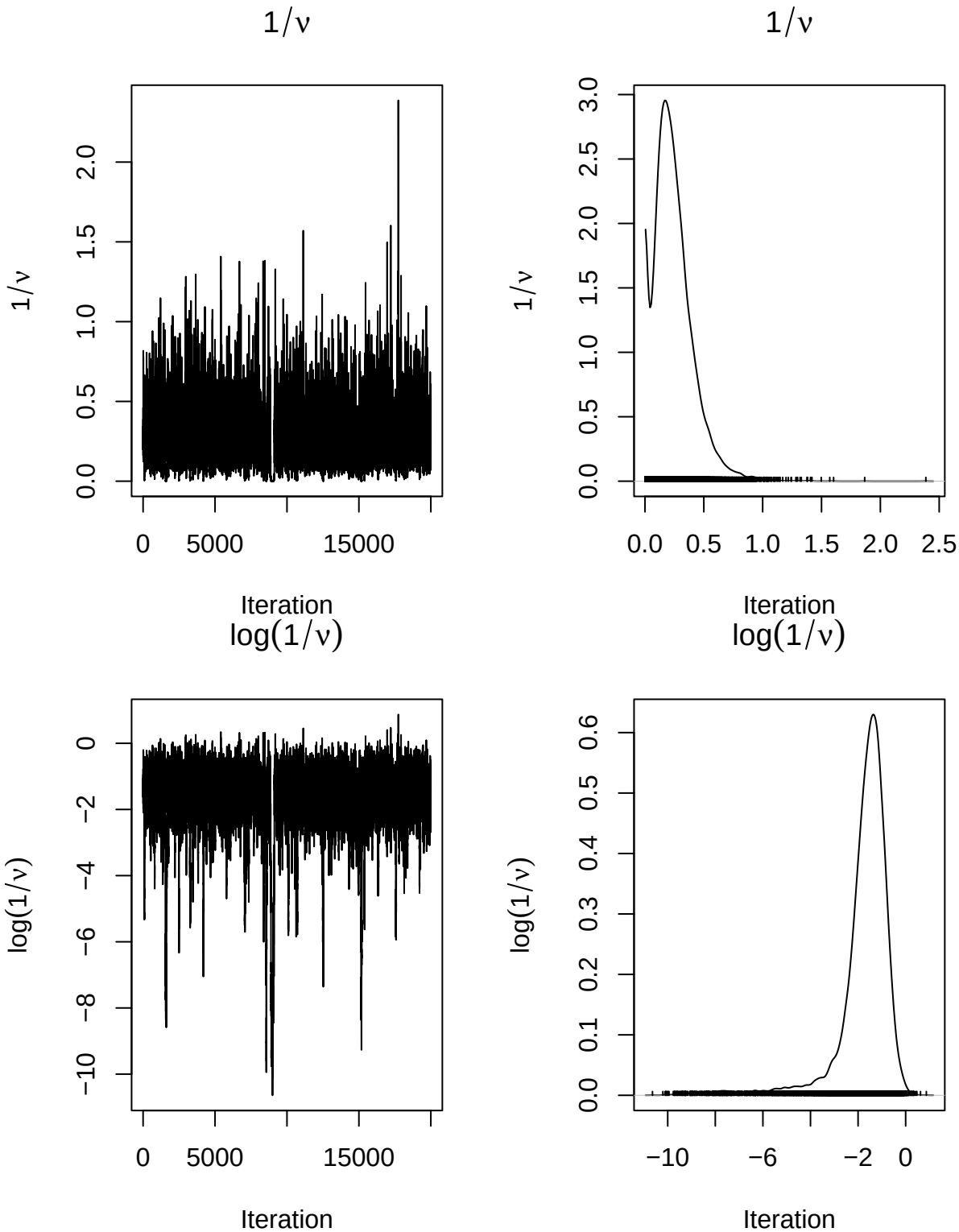
```
##           mu      gamma_1      gamma_2      gamma_3      gamma_4      gamma_5
## 10.186949480 -0.228328265 -0.241998135 -0.308130141  0.445724338  0.001276209
##      gamma_6      gamma_7      gamma_8      gamma_9     gamma_10     gamma_11
## -0.195878016  0.081798904  0.209191417  0.310980433  0.052458426  0.112042491
##     gamma_12     gamma_13     gamma_14     gamma_15     gamma_16     gamma_17
## -0.069377360 -0.319505738 -0.513020222  0.656524132 -0.441780123 -0.323819534
##     gamma_18     gamma_19     gamma_20   sigma2_eps sigma2_gamma
##  0.752023533  0.063353793 -0.013827620  0.802720740  0.246525959
```

The Gibbs sampler was run for 20000 iterations to sample from the joint posterior distribution of the model parameters. The posterior mean of the global intercept is approximately $\mu \approx 10.19$, which is very close to the maximum likelihood estimate obtained in part (a). The posterior mean of the residual variance is approximately $1/\tau \approx 0.80$, and the posterior mean of the random intercept variance is approximately $1/\nu \approx 0.25$. These values are close to the corresponding maximum likelihood estimates from part (a), namely

0.756 and 0.2301. This suggests that the Gibbs sampler is producing reasonable posterior estimates and that the implementation appears to be correct.

d)





The trace plots show that the Markov chains fluctuate around stable levels with no visible trends, suggesting that the Gibbs sampler mixes reasonably well. The posterior mean of the global intercept μ is around 10.2, which is very close to the maximum likelihood estimate obtained in part (a). The trace plot for μ shows tight fluctuations around this level and the posterior density appears approximately normal.

The residual variance $1/\tau$ fluctuates around approximately 0.8 and also appears stable across iterations. Its posterior distribution is slightly right-skewed, which is typical for variance parameters.

The random intercept variance $1/\nu$ behaves somewhat differently. Its trace plot shows larger variability and occasional spikes toward larger values. The posterior density is strongly right-skewed, with most of its mass concentrated near small values. This indicates that there is greater uncertainty in the random intercept variance than in the residual variance.

Examining the trace plot of $\log(1/\nu)$ helps visualize this behavior more clearly. On the log scale the chain appears more regular, while still showing occasional excursions corresponding to large values of $1/\nu$. Overall, the trace plots suggest that the Gibbs sampler mixes reasonably well, although the random intercept variance is clearly the most variable parameter in the model.

e)

From (6) we can rewrite

$$p = \frac{\nu^{-1}}{\nu^{-1} + \tau^{-1}} = \frac{\tau}{\tau + \nu}.$$

Under the prior in (4), the precision parameters are independent with

$$\tau \sim \Gamma(\alpha, \alpha), \qquad \nu \sim \Gamma(\alpha, \alpha).$$

A standard result for independent Gamma variables with common rate parameter states that if

$$X \sim \Gamma(a, \lambda), \qquad Y \sim \Gamma(b, \lambda),$$

then

$$\frac{X}{X+Y} \sim \text{Beta}(a, b).$$

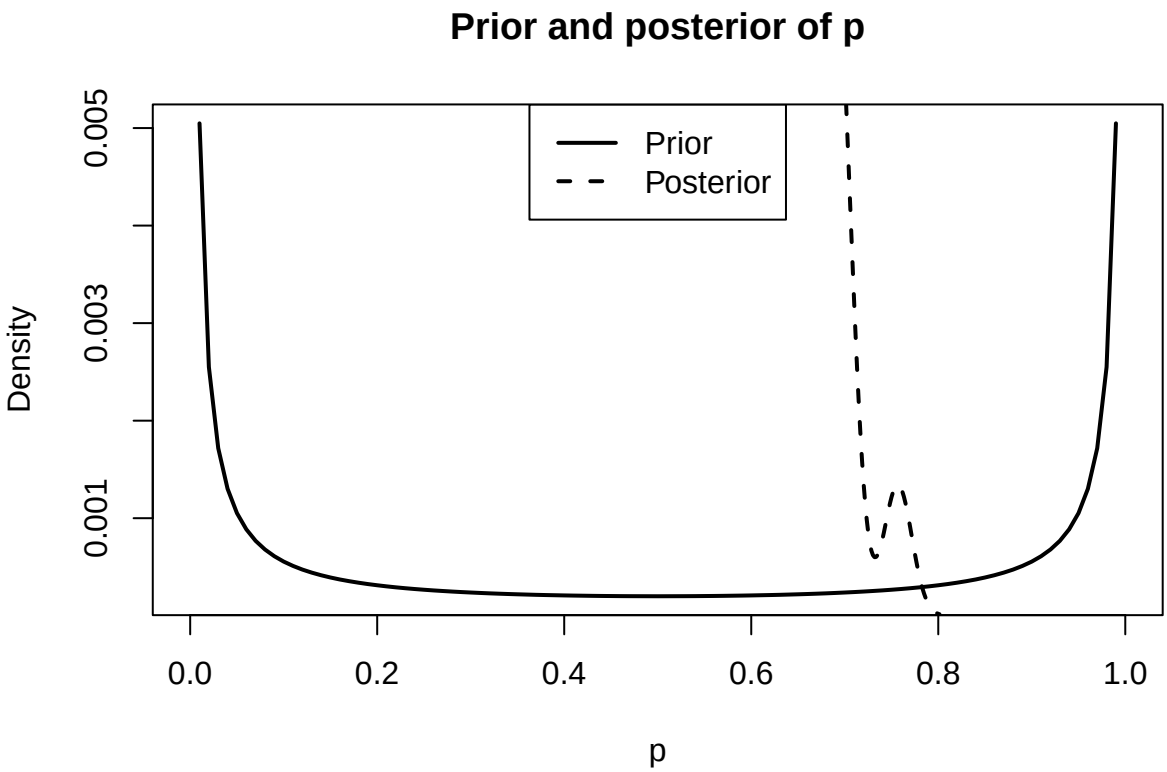
Applying this with $X = \tau$ and $Y = \nu$, we obtain

$$p = \frac{\tau}{\tau + \nu} \sim \text{Beta}(\alpha, \alpha).$$

Hence, with $\alpha = 0.0001$, the implied prior distribution is

$$p \sim \text{Beta}(0.0001, 0.0001).$$

This prior is strongly U-shaped, with most of its mass concentrated near 0 and 1. Thus it expresses a prior belief that the random intercept explains either almost none or almost all of the total variance. In a typical applied setting, this would usually not represent a very reasonable prior belief about p.



From part (4) we have independent priors

$$\tau \sim \Gamma(\alpha, \alpha), \qquad \nu \sim \Gamma(\alpha, \alpha).$$

Using the definition

$$p = \frac{\nu^{-1}}{\nu^{-1} + \tau^{-1}} = \frac{\tau}{\tau + \nu},$$

and the standard result that if

$$X \sim \Gamma(a, \lambda), \quad Y \sim \Gamma(b, \lambda),$$

independently, then

$$\frac{X}{X+Y} \sim \text{Beta}(a, b),$$

it follows that

$$p \sim \text{Beta}(\alpha, \alpha).$$

With $\alpha = 0.0001$, the prior for p is therefore

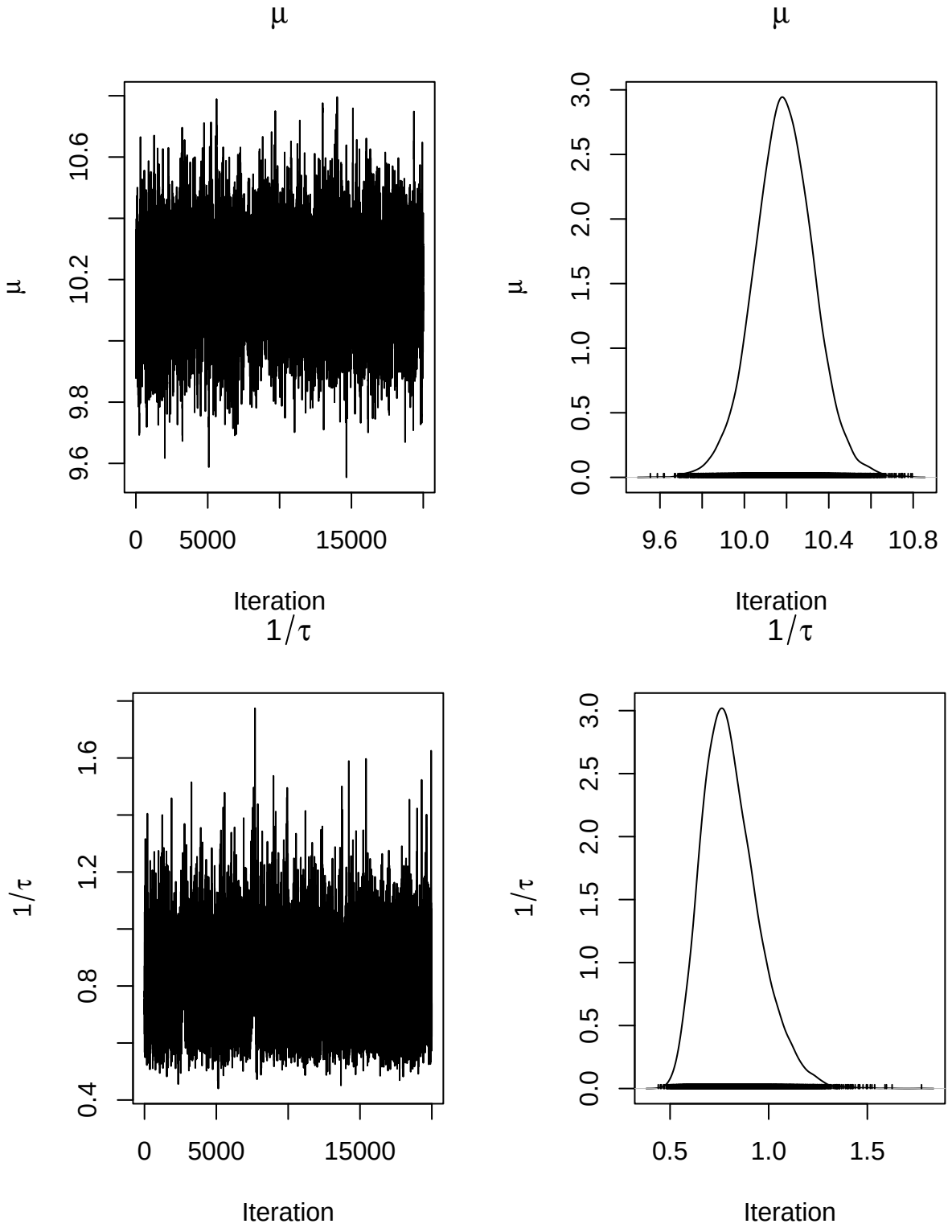
$$p \sim \text{Beta}(0.0001, 0.0001).$$

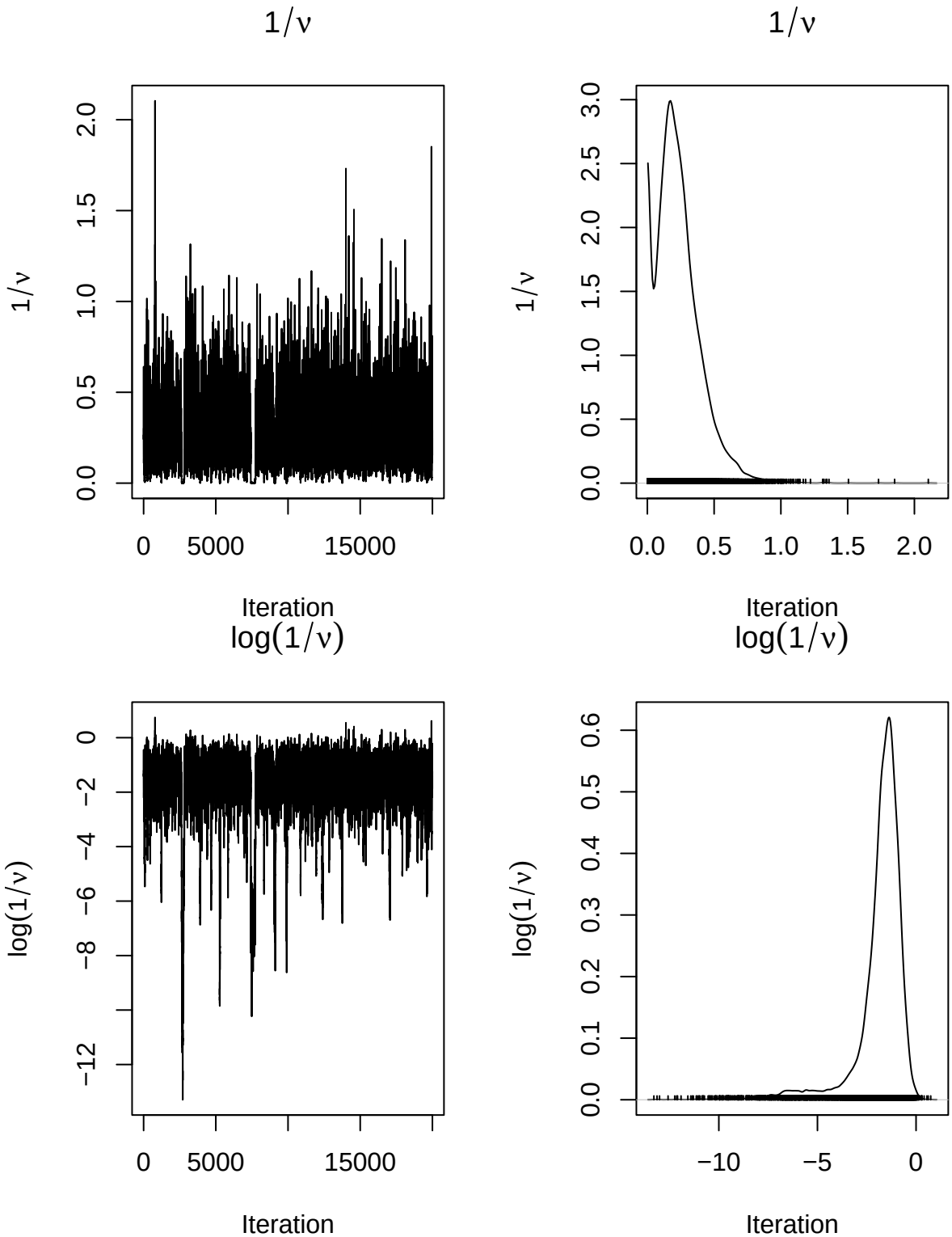
This prior distribution is strongly U-shaped and places most of its mass near 0 and 1. Hence it expresses the belief that the random intercept explains either almost none or almost all of the total variance.

The posterior distribution obtained from the Gibbs sampler is concentrated around values near $p \approx 0.7$, indicating that the data suggest a substantial but not dominant contribution of the random intercept to the total variance. The posterior is much more concentrated than the prior and lies far from the extreme values favored by the prior.

Therefore, the prior implied by (4) does not appear to represent a very reasonable prior belief about p in a typical applied setting.

f)


μ
μ
10.6
10.2
9.8
9.6
0
5000
15000
Iteration
μ
3.0
2.5
2.0
1.5
1.0
0.5
0.0
9.6
10.0
10.4
10.8
Iteration
1/τ
1/τ
1.6
1.2
0.8
0.4
0
5000
15000
Iteration
1/τ
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.5
1.0
1.5
Iteration



In the limiting case $\alpha \to 0$, the Gamma priors in (4) become the improper scale priors

$$\pi(\tau) \propto \frac{1}{\tau}, \qquad \pi(\nu) \propto \frac{1}{\nu}.$$

Running the Gibbs sampler with $\alpha = 0$, the chains for the global intercept μ and the residual variance

$1/\tau$ remain stable and well-behaved. Their trace plots fluctuate around constant levels and their posterior distributions appear similar to those obtained in part (d).

The behavior of the random intercept variance $1/\nu$ is quite different. The trace plot shows much larger variability with occasional large spikes. The posterior density is strongly right-skewed, indicating that very large values of the variance component occur with non-negligible probability.

Examining the trace plot of $\log(1/\nu)$ further highlights this behavior, showing large excursions corresponding to extreme values of the random intercept variance. This suggests that the posterior for ν is weakly constrained under the improper prior $\pi(\nu) \propto 1/\nu$.

Although the full conditional distributions remain proper, this behavior indicates that the joint posterior may be close to improper or poorly behaved for this choice of prior. The variance component is therefore much less stable under the improper scale prior than under the Gamma prior used in part (d).

g)

Define

$$\sigma^2 = \nu^{-1} + \tau^{-1}, \qquad p = \frac{\nu^{-1}}{\nu^{-1} + \tau^{-1}}.$$

Then

$$\nu^{-1} = p\sigma^2, \qquad \tau^{-1} = (1-p)\sigma^2,$$

so that

$$\nu = \frac{1}{p\sigma^2}, \qquad \tau = \frac{1}{(1-p)\sigma^2}.$$

To derive the implied prior on (ν, τ), it is convenient to write

$$\sigma^2 = \frac{1}{\nu} + \frac{1}{\tau} = \frac{\nu+\tau}{\nu\tau}, \qquad p = \frac{\nu^{-1}}{\nu^{-1} + \tau^{-1}} = \frac{\tau}{\nu+\tau}.$$

The Jacobian of the transformation $(\nu, \tau) \mapsto (\sigma^2, p)$ is

$$\frac{\partial(\sigma^2, p)}{\partial(\nu, \tau)} = \begin{pmatrix} -\nu^{-2} & -\tau^{-2} \\ -\dfrac{\tau}{(\nu+\tau)^2} & \dfrac{\nu}{(\nu+\tau)^2} \end{pmatrix},$$

with determinant

$$\left| \frac{\partial(\sigma^2, p)}{\partial(\nu, \tau)} \right| = \frac{1}{\nu\tau(\nu+\tau)}.$$

Since the prior in (7a) is

$$\pi(\sigma^2, p) \propto \frac{1}{\sigma^2} I(0 < p < 1) I(\sigma^2 > 0),$$

and

$$\frac{1}{\sigma^2} = \frac{\nu\tau}{\nu+\tau},$$

it follows by change of variables that

$$\pi(\nu, \tau) \propto \frac{\nu\tau}{\nu+\tau} \cdot \frac{1}{\nu\tau(\nu+\tau)} = \frac{1}{(\nu+\tau)^2},$$

for $\nu > 0$ and $\tau > 0$. Hence

$$\pi(\nu, \tau) \propto \frac{1}{(\nu+\tau)^2} I(\nu > 0) I(\tau > 0).$$

Next, fixing ν, the conditional prior density of τ is

$$\pi(\tau \mid \nu) \propto \frac{1}{(\nu+\tau)^2}, \qquad \tau > 0.$$

Let $x = \ln \tau$, so that $\tau = e^x$ and $d\tau/dx = e^x$. Then

$$\pi(x \mid \nu) \propto \frac{e^x}{(\nu + e^x)^2} = \frac{e^{x - \ln \nu}}{(1 + e^{x - \ln \nu})^2}.$$

This is the density of a logistic distribution with location parameter $\ln \nu$ and scale 1. Therefore,

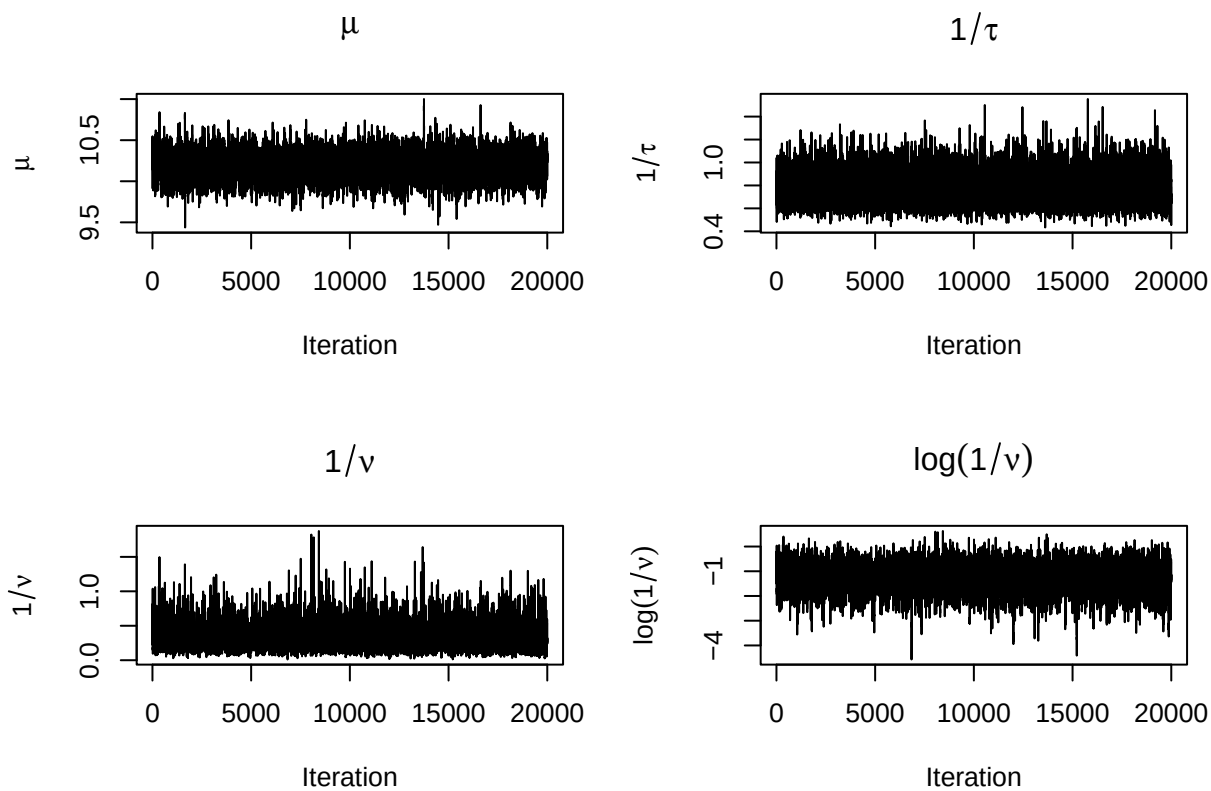
$$\ln \tau \mid \ln \nu \sim \text{Logistic}(\ln \nu, 1).$$

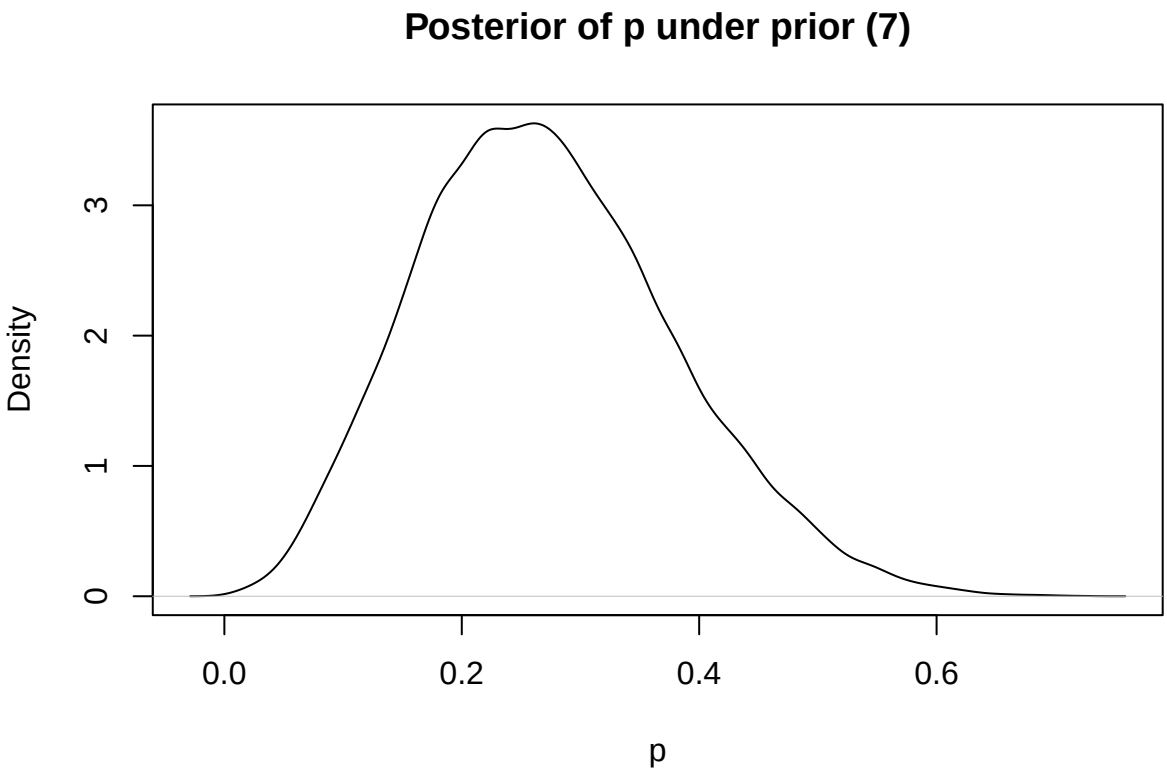
..

h)

```
## [1] 0.96535
```

```
## [1] 0.9189
```





To sample from the posterior under the prior in (7), I implemented a Metropolis-within-Gibbs sampler. The updates for μ and γ_i remain unchanged from part (c), since their full conditional distributions are still normal. However, under the prior

$$\pi(\nu, \tau) \propto \frac{1}{(\nu + \tau)^2},$$

the full conditional distributions for τ and ν are no longer Gamma distributions. Therefore, these parameters were updated using Metropolis–Hastings steps.

As proposal distributions, I used the Gamma full conditionals derived in part (f):

$$\tau' \sim \Gamma\left(\frac{N}{2}, \frac{1}{2}\sum(y - \mu - \gamma_i)^2\right), \qquad \nu' \sim \Gamma\left(\frac{m}{2}, \frac{1}{2}\sum \gamma_i^2\right).$$

This results in acceptance probabilities

$$A_\tau = \min\left(1, \frac{\tau'}{(\nu + \tau')^2}\frac{(\nu + \tau)^2}{\tau}\right),$$

and

$$A_\nu = \min\left(1, \frac{\nu'}{(\nu' + \tau)^2}\frac{(\nu + \tau)^2}{\nu}\right).$$

Running the sampler for 20000 iterations produced acceptance rates of approximately 0.97 for τ and 0.92 for ν, indicating that the proposal distributions lead to very efficient updates.

The trace plots show that the chains for μ and $1/\tau$ remain stable and well-behaved, similar to the earlier results. Importantly, the behavior of the random intercept variance $1/\nu$ is now considerably more stable than in part (f). The trace plot shows smaller fluctuations and no extreme excursions, and the trace of $\log(1/\nu)$ appears well-mixed without large spikes. This indicates improved convergence properties under the new prior.

The posterior distribution of the variance proportion

$$p = \frac{\nu^{-1}}{\nu^{-1} + \tau^{-1}}$$

is concentrated around moderate values, suggesting that a substantial but not dominant fraction of the total variance is explained by the random intercept. Unlike the earlier prior, the prior in (7) induces a uniform prior on p, avoiding the strong concentration near 0 and 1 that occurred under the Gamma prior in part (e). Consequently, the posterior under this prior appears more stable and interpretable.

Overall, the Gibbs sampler with the prior in (7) shows better convergence properties than the sampler using the improper scale prior in part (f), particularly for the random intercept variance component.

Problem 3

Problem 4

We consider the state equation

$$u_t = u_{t-1} + r\left(1 - \left(\frac{e^{u_{t-1}}}{K}\right)^{\theta}\right) + e_t, \qquad e_t \sim \mathcal{N}(0, \sigma), \tag{1}$$

and the observation equation

$$y_t \mid u_t \sim \text{Poisson}(ce^{u_t}). \tag{2}$$

a)

To show that the model is non-identifiable, it suffices to demonstrate that different parameter values yield the same distribution for the observations $y = (y_1, \ldots, y_n)$.

Let $a \in \mathbb{R}$ and define

$$u_t^* = u_t + a, \qquad K^* = Ke^a, \qquad c^* = ce^{-a},$$

with

$$r^* = r, \qquad \theta^* = \theta, \qquad \sigma^* = \sigma.$$

Then

$$\frac{e^{u_{t-1}^*}}{K^*} = \frac{e^{u_{t-1}+a}}{Ke^a} = \frac{e^{u_{t-1}}}{K},$$

so the state equation remains unchanged:

$$u_t^* = u_{t-1}^* + r\left(1 - \left(\frac{e^{u_{t-1}^*}}{K^*}\right)^{\theta}\right) + e_t.$$

For the observation model,

$$c^* e^{u_t^*} = (ce^{-a})e^{u_t+a} = ce^{u_t},$$

so the conditional distribution of y_t is unchanged.

Since u_1 has an improper uniform distribution on $\mathbb{R}$, it is invariant under shifts. Hence, the transformation

$$(u_t, K, c) \mapsto (u_t + a, Ke^a, ce^{-a})$$

leaves the likelihood unchanged, and the model is non-identifiable.

Identifiability. The non-identifiability arises from a shift invariance in u_t. The quantity

$$\log K + \log c$$

is invariant under the transformation, implying that only the product cK is identifiable. To remove this invariance, we impose the constraint

$$c = 1,$$

which yields an identifiable model without altering its structure.

b)

To resolve the non-identifiability shown in part (a), we fix $c = 1$. The parameters r, K, θ, σ are estimated, with K, θ, and σ optimized on the log-scale to enforce positivity.

We write the negative joint log-likelihood as

$$-\log f(y, u \mid \theta) = -\sum_{t=1}^{n} \log p(y_t \mid u_t) - \sum_{t=2}^{n} \log p(u_t \mid u_{t-1}),$$

where

Observation model

$$y_t \mid u_t \sim \text{Poisson}(e^{u_t}),$$

so

$$-\log p(y_t \mid u_t) = e^{u_t} - y_t u_t + \log(y_t!).$$

State model

$$u_t \mid u_{t-1} \sim \mathcal{N}(\mu_t, \sigma^2), \quad \mu_t = u_{t-1} + r\left(1 - \left(\frac{e^{u_{t-1}}}{K}\right)^{\theta}\right),$$

giving

$$-\log p(u_t \mid u_{t-1}) = \frac{(u_t - \mu_t)^2}{2\sigma^2} + \log \sigma + \text{const}.$$

The latent states $u = (u_1, \ldots, u_n)$ are treated as random effects in `RTMB`. The marginal likelihood is approximated using the Laplace approximation via `MakeADFun` with `random = "u"`. Parameters K, θ, and σ are optimized on the log-scale to enforce positivity, and $c = 1$ is fixed to ensure identifiability.

c)

The Laplace approximation of the marginal likelihood was maximised using the `nlminb` optimizer. The optimisation converged successfully, with convergence code 0 and a message indicating both parameter and relative convergence. The maximum absolute gradient at the optimum was $8.96 \cdot 10^{-7}$, which is close to zero and confirms that a stationary point was reached.

```
## [1] 0
```

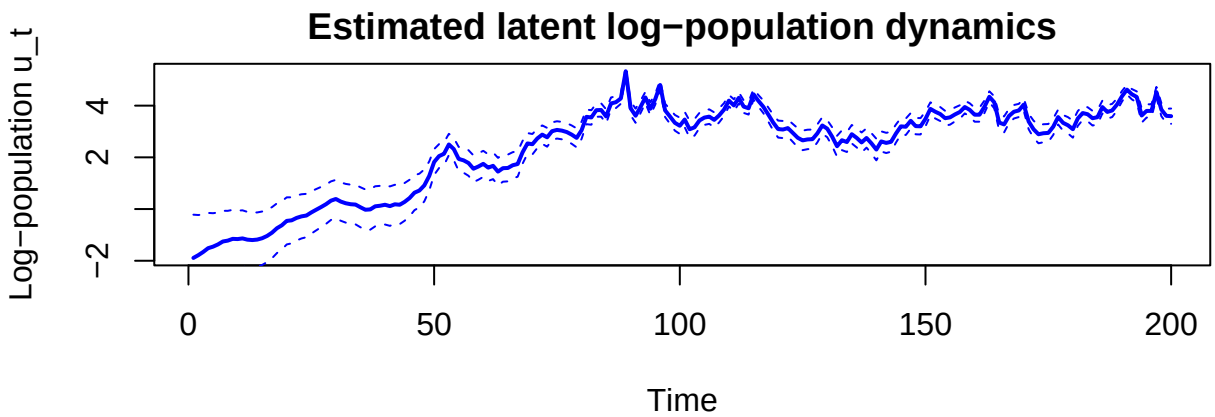
```
## [1] "both X-convergence and relative convergence (5)"
```

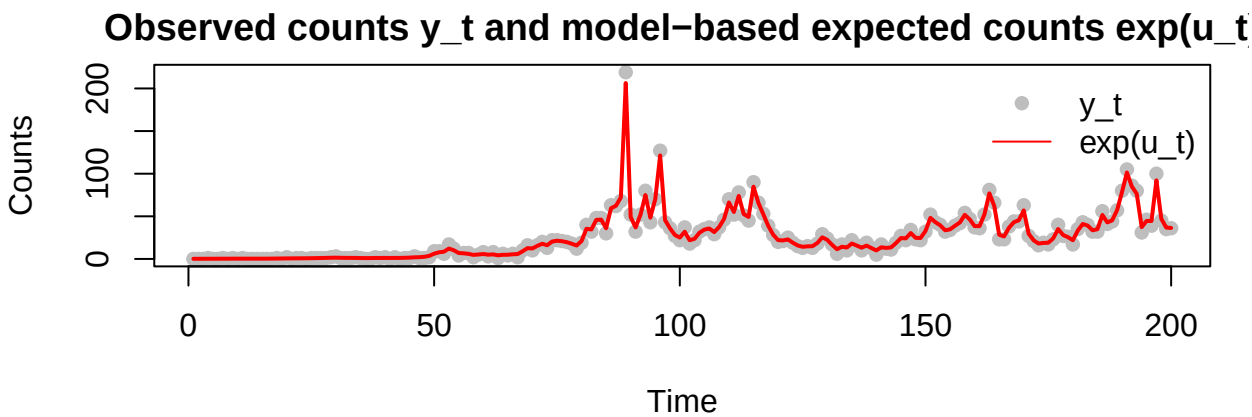
```
## [1] 637.1859
```

```
## iter: 1  mgc: 2.984279e-13
## outer mgc:  8.964445e-07
```

```
## [1] 8.964445e-07
```

Parameter estimates and standard errors were obtained using `sdreport`, which is based on a Gaussian approximation around the optimum using the curvature of the objective function. In particular, `sdreport` provides both the estimated latent states and their uncertainties.





The estimated latent states $\hat{u}_t$ are shown in the first plot. Since u_t represents the latent log-population, this plot describes the model's estimate of the underlying population dynamics on the log-scale. The trajectory is smooth over time because the latent states are constrained by the theta-logistic state model, and the confidence bands are relatively narrow, suggesting that the latent process is well informed by the data.

The second plot compares the observed counts y_t with the fitted values $\exp(\hat{u}_t)$, which represent the model-based expected counts on the original scale. The fitted curve follows the observed data closely, including periods of rapid increase and decline, showing that the model captures the main structure of the data well. At the same time, the fitted values are not obtained by fitting each time point independently, but through the latent state process, which imposes temporal dependence and yields a coherent trajectory over time.

Overall, the model provides a good fit and successfully recovers an underlying latent population process from noisy count observations.

d)

To evaluate the performance of the method, we simulated a dataset of $n = 200$ observations using known parameter values and fitted the model to the simulated data.

Results

The parameter estimates are close to the true values, and all true parameters lie within the corresponding 95% confidence intervals. This indicates that the estimation procedure performs well and that the Laplace approximation provides reliable inference for this model.

The parameter θ exhibits the largest uncertainty, reflected in a relatively wide confidence interval. This is expected, as θ governs the nonlinear density dependence and enters the model in a highly nonlinear manner, making it more difficult to estimate from finite data. In contrast, σ is estimated with high precision, indicating

Table 1: Comparison of true parameter values and estimates based on simulated data.

	Parameter	True value	Estimate	Std. error	95% CI
r	r	0.1	0.1134	0.0431	[0.029, 0.198]
K	K	45.0	43.2630	4.8223	[33.811, 52.715]
theta	θ	2.0	2.1603	0.7618	[0.667, 3.654]
sigma	σ	0.3	0.2718	0.0211	[0.23, 0.313]

that the noise level is well identified. The parameters r and K are also reasonably well estimated, suggesting that the overall growth dynamics are captured effectively.

The results are based on a single simulated dataset, and the estimation performance may vary across different realizations. In addition, the Laplace approximation relies on a Gaussian approximation of the latent posterior distribution, which may be less accurate in highly nonlinear settings or for small sample sizes. Nevertheless, the results suggest that the approximation performs well in this setting.

Fixing $c = 1$ was essential to ensure identifiability. Without this constraint, the model exhibits a shift invariance in the latent states, making it impossible to uniquely determine the parameters. Imposing this constraint leads to a well-defined estimation problem and stable numerical optimization.

Standard errors

The standard errors reported by `sdreport` are based on asymptotic likelihood theory. The covariance matrix is approximated by the inverse Hessian of the Laplace-approximated marginal negative log-likelihood,

$$H(\hat{\theta})^{-1}.$$

Since parameters are optimized on the log-scale, the Delta method is used to obtain standard errors on the original scale:

$$\mathrm{Var}(g(\theta)) \approx \nabla g(\theta)^T H(\theta)^{-1} \nabla g(\theta).$$

Overall, the standard errors appear to be of a reasonable magnitude, and the agreement between estimated and true parameter values provides empirical support for the reliability of the Laplace approximation in this model.